Proxy gaming more generally. In these examples, the systems are given an approximate—"proxy"—goal or objective that initially seems to correlate with the ideal goal. However, they end up exploiting this proxy in ways that diverge from the idealized goal or even lead to negative outcomes. Offering a reward for rat tails seems like a good way to reduce the population of rats; a patient's healthcare costs appear to be an accurate indication of health risk; and a boat race reward system should encourage boats to race, not catch themselves on fire. Yet, in each instance, the system optimized its proxy objective in ways that did not achieve the intended outcome or even made things worse overall. This phenomenon is captured by Goodhart's law: "Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes," or put succinctly but overly simplistically, "when a measure becomes a target, it ceases to be a good measure." In other words, there may usually be a statistical regularity between healthcare costs and poor health, or between targets hit and finishing the course, but when we place pressure on it by using one as a proxy for the other, that relationship will tend to collapse.

Correctly specifying goals is no trivial task. If delineating exactly what we want from a boat racing AI is tricky, capturing the nuances of human values under all possible scenarios will be much harder. Philosophers have been attempting to precisely describe morality and human values for millennia, so a precise and flawless characterization is not within reach. Although we can refine the goals we give AIs, we might always rely on proxies that are easily definable and measurable. Discrepancies between the proxy goal and the intended function arise for many reasons. Besides the difficulty of exhaustively specifying everything we care about, there are also limits to how much we can oversee AIs, in terms of time, computational resources, and the number of aspects of a system that can be monitored. Additionally, AIs may not be adaptive to new circumstances or robust to adversarial attacks that seek to misdirect them. As long as we give AIs proxy goals, there is the chance that they will find loopholes we have not thought of, and thus find unexpected solutions that fail to pursue the ideal goal.

The more intelligent an AI is, the better it will be at gaming proxy goals. Increasingly intelligent agents can be increasingly capable of finding unanticipated routes to optimizing proxy goals without achieving the desired outcome [118]. Additionally, as we grant AIs more power to take actions in society, for example by using them to automate certain processes, they will have access to more means of achieving their goals. They may then do this in the most efficient way available to them, potentially causing harm in the process. In a worst case scenario, we can imagine a highly powerful agent optimizing a flawed objective to an extreme degree without regard for human life. This represents a catastrophic risk of proxy gaming.

In summary, it is often not feasible to perfectly define exactly what we want from a system, meaning that many systems find ways to achieve their given goal without performing their intended function. AIs have already been observed to do this, and are likely to get better at it as their capabilities improve. This is one possible mechanism that could result in an uncontrolled AI that would behave in unanticipated and potentially harmful ways.

5.2 Goal Drift

Even if we successfully control early AIs and direct them to promote human values, future AIs could end up with different goals that humans would not endorse. This process, termed "goal drift," can be hard to predict or control. This section is most cutting-edge and the most speculative, and in it we will discuss how goals shift in various agents and groups and explore the possibility of this phenomenon occurring in AIs. We will also examine a mechanism that could lead to unexpected goal drift, called intrinsification, and discuss how goal drift in AIs could be catastrophic.

The goals of individual humans change over the course of our lifetimes. Any individual reflecting on their own life to date will probably find that they have some desires now that they did not have earlier in their life. Similarly, they will probably have lost some desires that they used to have. While we may be born with a range of basic desires, including for food, warmth, and human contact, we develop many more over our

lifetime. The specific types of food we enjoy, the genres of music we like, the people we care most about, and the sports teams we support all seem heavily dependent on the environment we grow up in, and can also change many times throughout our lives. A concern is that individual AI agents may have their goals change in complex and unanticipated ways, too.

Groups can also acquire and lose collective goals over time. Values within society have changed throughout history, and not always for the better. The rise of the Nazi regime in 1930s Germany, for instance, represented a profound moral regression, which ultimately resulted in the systematic extermination of six million Jews during the Holocaust, alongside widespread persecution of other minority groups. Additionally, the regime greatly restricted freedom of speech and expression. Here, a society's goals drifted for the worse.

The Red Scare that took place in the United States from 1947-1957 is another example of societal values drifting. Fuelled by strong anti-communist sentiment, against the backdrop of the Cold War, this period saw the curtailment of civil liberties, widespread surveillance, unwarranted arrests, and blacklisting of suspected communist sympathizers. This constituted a regression in terms of freedom of thought, freedom of speech, and due process. Just as the goals of human collectives can change in emergent and unexpected ways, collectives of AI agents may also have their goals unexpectedly drift from the ones we initially gave them.

Over time, instrumental goals can become intrinsic. Intrinsic goals are things we want for their own sake, while instrumental goals are things we want because they can help us get something else. We might have an intrinsic desire to spend time on our hobbies, simply because we enjoy them, or to buy a painting because we find it beautiful. Money, meanwhile, is often cited as an instrumental desire; we want it because it can buy us other things. Cars are another example; we want them because they offer a convenient way of getting around. However, an instrumental goal can become an intrinsic one, through a process called intrinsification. Since having more money usually gives a person greater capacity to obtain things they want, people often develop a goal of acquiring more money, even if there is nothing specific they want to spend it on. Although people do not begin life desiring money, experimental evidence suggests that receiving money can activate the reward system in the brains of adults in the same way that pleasant tastes or smells do [119, 120]. In other words, what started as a means to an end can become an end in itself.

This may happen because the fulfillment of an intrinsic goal, such as purchasing a desired item, produces a positive reward signal in the brain. Since having money usually coincides with this positive experience, the brain associates the two, and this connection will strengthen to a point where acquiring money alone can stimulate the reward signal, regardless of whether one buys anything with it [121].

It is feasible that intrinsification could happen with AI agents. We can draw some parallels between how humans learn and the technique of reinforcement learning. Just as the human brain learns which actions and conditions result in pleasure and which cause pain, AI models that are trained through reinforcement learning identify which behaviors optimize a reward function, and then repeat those behaviors. It is possible that certain conditions will frequently coincide with AI models achieving their goals. They might, therefore, intrinsify the goal of seeking out those conditions, even if that was not their original aim.

AIs that intrinsify unintended goals would be dangerous. Since we might be unable to predict or control the goals that individual agents acquire through intrinsification, we cannot guarantee that all their acquired goals will be beneficial for humans. An originally loyal agent could, therefore, start to pursue a new goal without regard for human wellbeing. If such a rogue AI had enough power to do this efficiently, it could be highly dangerous.

AIs will be adaptive, enabling goal drift to happen. It is worth noting that these processes of drifting goals are possible if agents can continually adapt to their environments, rather than being essentially "fixed" after the training phase. Indeed, this adaptability is the likely reality we face. If we want AIs to complete the tasks we assign them effectively and to get better over time, they will need to be adaptive, rather than set in stone.

They will be updated over time to incorporate new information, and new ones will be created with different designs and datasets. However, adaptability can also allow their goals to change.

If we integrate an ecosystem of agents in society, we will be highly vulnerable to their goals drifting. In a potential future scenario where AIs have been put in charge of various decisions and processes, they will form a complex system of interacting agents. A wide range of dynamics could develop in this environment. Agents might imitate each other, for instance, creating feedback loops, or their interactions could lead them to collectively develop unanticipated emergent goals. Competitive pressures may also select for agents with certain goals over time, making some initial goals less represented compared to fitter goals. These processes make the long-term trajectories of such an ecosystem difficult to predict, let alone control. If this system of agents were enmeshed in society and we were largely dependent on them, and if they gained new goals that superseded the aim of improving human wellbeing, this could be an existential risk.

5.3 Power-Seeking

So far, we have considered how we might lose our ability to control the goals that AIs pursue. However, even if an agent started working to achieve an unintended goal, this would not necessarily be a problem, as long as we had enough power to prevent any harmful actions it wanted to attempt. Therefore, another important way in which we might lose control of AIs is if they start trying to obtain more power, potentially transcending our own. We will now discuss how and why AIs might become power-seeking and how this could be catastrophic. This section draws heavily from "Existential Risk from Power-Seeking AI" [122].

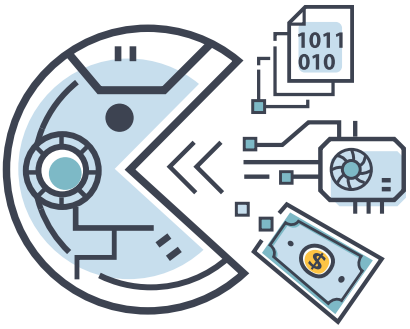

Figure 16: Various resources, such as money and computing power, can sometimes be instrumentally rational to seek. AIs which can capably pursue goals may take intermediate steps to gain power and resources.

AIs might seek to increase their own power as an instrumental goal. In a scenario where rogue AIs were pursuing unintended goals, the amount of damage they could do would hinge on how much power they had. This may not be determined solely by how much control we initially give them; agents might try to get more power, through legitimate means, deception, or force. While the idea of power-seeking often evokes an image of "power-hungry" people pursuing it for its own sake, power is often simply an instrumental goal. The ability to control one's environment can be useful for a wide range of purposes: good, bad, and neutral. Even if an individual's only goal is simply self-preservation, if they are at risk of being attacked by others, and if they cannot rely on others to retaliate against attackers, then it often makes sense to seek power to help avoid being harmed—no *animus dominandi* or lust for power is required for power-seeking behavior to emerge [123]. In other words, the environment can make power acquisition instrumentally rational.

AIs trained through reinforcement learning have already developed instrumental goals including tool-use. In one example from OpenAI, agents were trained to play hide and seek in an environment with various objects scattered around [124]. As training progressed, the agents tasked with hiding learned to use these objects to construct shelters around themselves and stay hidden. There was no direct reward for this tool-use behavior; the hiders only received a reward for evading the seekers, and the seekers only for finding the hiders. Yet they learned to use tools as an instrumental goal, which made them more powerful.

Self-preservation could be instrumentally rational even for the most trivial tasks. An example by computer scientist Stuart Russell illustrates the potential for instrumental goals to emerge in a wide range of AI systems [125]. Suppose we tasked an agent with fetching coffee for us. This may seem relatively harmless, but